

Xingyu Su

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RESEARCH INTEREST

My research focuses on generative modeling and optimization, with an emphasis on **discrete diffusion models**, **reinforcement learning methods**, and **large language models**. I am the author of **Seq2Exp**, a framework captures causal relationships between DNA and epigenomic data for gene-expression prediction, selected as an oral paper at ICLR 2025. I also proposed **VIDD** (Iterative Distillation for Reward-Guided Fine-Tuning of Diffusion Models in Biomolecular Design), which develops reinforcement learning methods to fine-tune discrete diffusion models for protein and DNA design.

EDUCATION

Texas A&M University <i>Ph.D. in Computer Science & Engineering</i> • Advisor: Prof. Shuiwang Ji	College Station, TX <i>Sep 2023 – Present</i>
Shanghai Jiao Tong University <i>Master of Science in Computer Science</i> • Advisor: Prof. Yanyan Shen	Shanghai, China <i>Sep 2020 – Feb 2023</i>
Shanghai Jiao Tong University <i>Bachelor of Science in Computer Science</i>	Shanghai, China <i>Sep 2016 – Aug 2020</i>

PUBLICATIONS

- **Xingyu Su***, Haiyang Yu*, Degui Zhi, and Shuiwang Ji. Learning to Discover Regulatory Elements for Gene Expression Prediction. International Conference on Learning Representations (ICLR 2025, oral 1.8%).
- Masatoshi Uehara*, **Xingyu Su***, Yulai Zhao, Xiner Li, Aviv Regev, Shuiwang Ji, Sergey Levine, Tommaso Biancalani. Reward-Guided Iterative Refinement in Diffusion Models at Test-Time with Applications to Protein and DNA Design. International Conference on Machine Learning (ICML 2025).
- **Xingyu Su***, Xiner Li*, Masatoshi Uehara*, Sunwoo Kim, Yulai Zhao, Gabriele Scalia, Ehsan Hajiramezanali, Tommaso Biancalani, Degui Zhi, Shuiwang Ji. Iterative Distillation for Reward-Guided Fine-Tuning of Diffusion Models in Biomolecular Design. arXiv preprint arXiv:2507.00445 (2025).
- Xiner Li*, Masatoshi Uehara*, **Xingyu Su**, Gabriele Scalia, Tommaso Biancalani, Aviv Regev, Sergey Levine, Shuiwang Ji. Dynamic Search for Inference-Time Alignment in Diffusion Models. arXiv preprint arXiv:2503.02039 (2025).
- **Xingyu Su**, Xiner Li, Yuchao Lin, Ziqian Xie, Degui Zhi, Shuiwang Ji. Language Models for Controllable DNA Sequence Design. arXiv preprint arXiv:2507.19523 (2025).

EXPERIENCE

The DIVE Lab, Texas A&M University <i>Graduate Student Researcher</i>	Sep 2023 – Present <i>College Station, TX</i>
Meituan <i>Recommendation Algorithm Engineer (Full-time)</i>	Feb 2023 – Jul 2023 <i>Shanghai, China</i>
Intel <i>Algorithm Engineer (Intern)</i>	Aug 2022 – Dec 2022 <i>Shanghai, China</i>
Shanghai Jiao Tong University <i>Teaching Assistant</i>	2019 <i>Shanghai, China</i>

AWARDS & HONORS

Graduate Research Excellence, TAMU CSE, one per year	2025
Undergraduate Merit Scholarship	2019
Fuda Scholarship	2019